

5-Mark Questions & Answers — Data Structures Module 2: Stacks & Queues

Q1. Write the complete algorithm and C functions for Infix to Postfix conversion using a Stack.

Detailed operator precedence algorithm handling `^` (highest, right-associative), `*`, `/`, `+`, `-`, and parentheses with complete C code.

Q2. Explain Circular Queue operations (Enqueue, Dequeue) with array implementation and boundary condition checks in C.

- **Enqueue condition:** `(rear + 1) % MAX == front` (Full).
- **Dequeue condition:** `front == -1` (Empty); resets `front = rear = -1` when last element is removed.

Q3. Explain how recursion is implemented using the Call Stack with an example (Factorial / Tower of Hanoi).

Step-by-step memory trace diagram showing Activation Records (Stack Frames) pushed during recursive calls and popped during return.